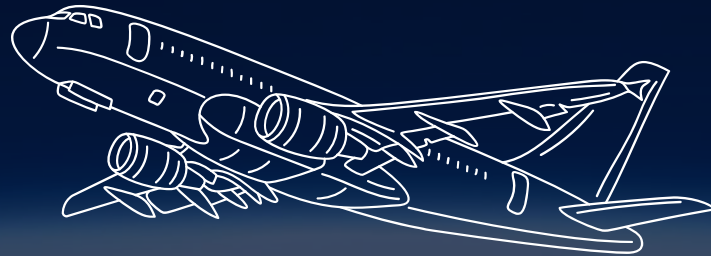


Principle of flight



By eng: Mohamed Ahmed El-shoura



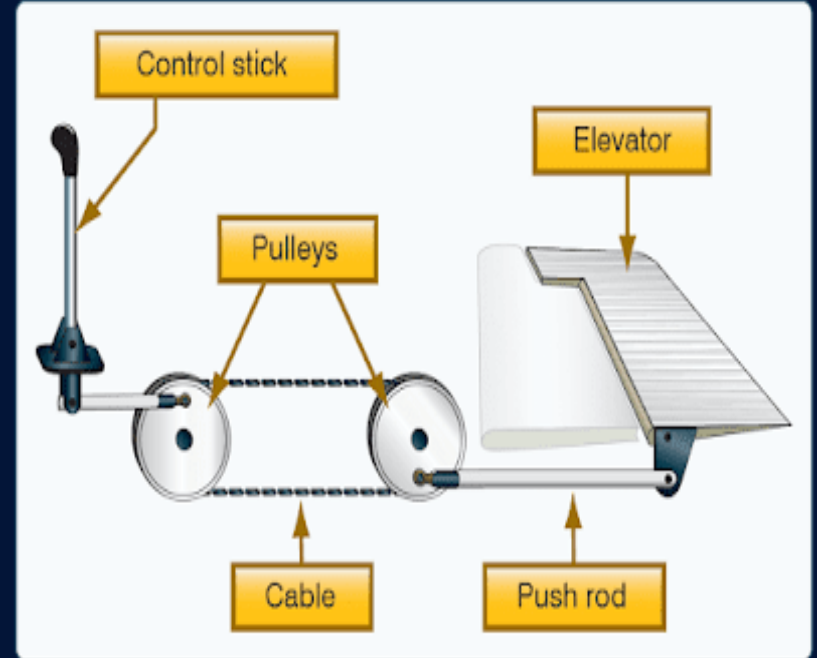
Flight Control Systems

Primary Flight Controls

Mainly used in small and low-speed aircrafts:

- The control forces required to give the control moments is sufficient to be handled directly by the pilot (adequate stick force)
- So, the primary control surface is not used in large and high-speed aircrafts as the control forces would be too high to be handled by the pilot.

$$L_{control} = C_L \frac{1}{2} \rho V^2 S$$

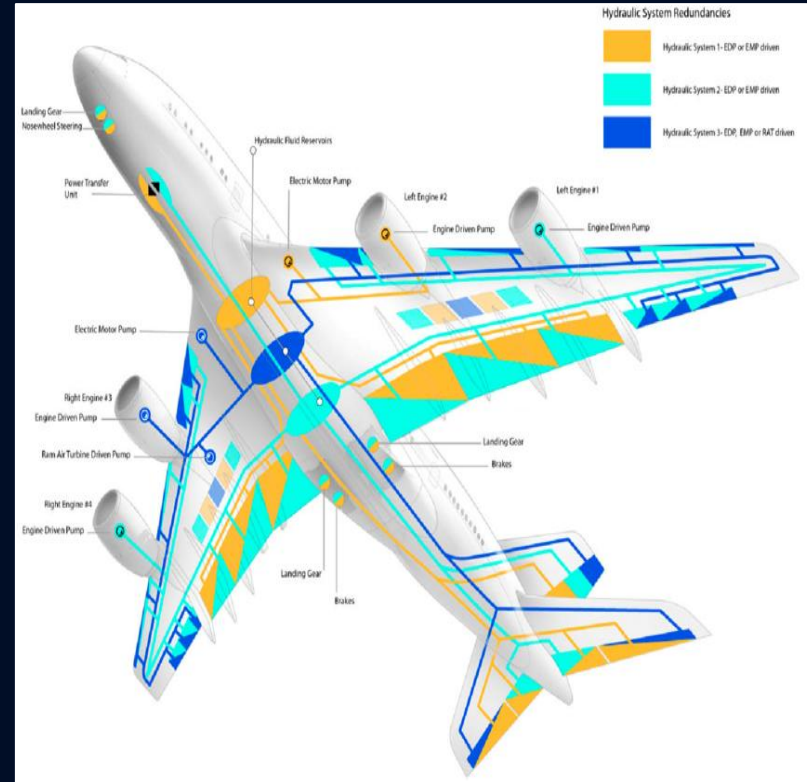


Powered Flight Controls

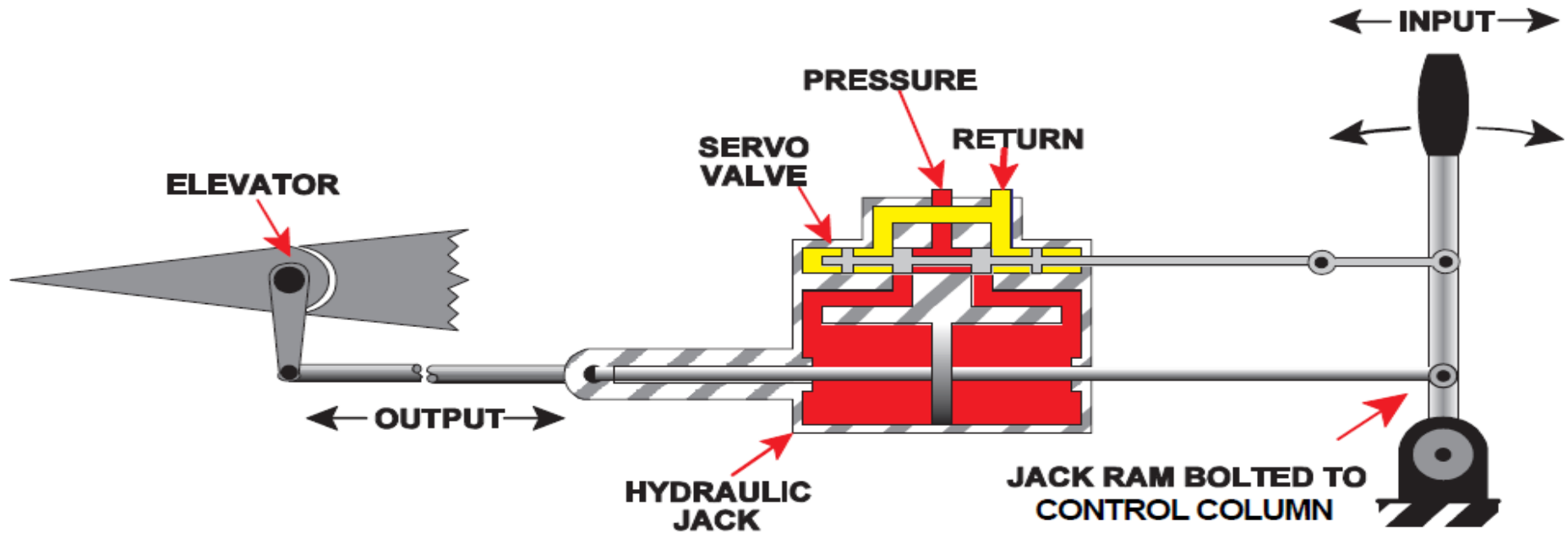
Used in large and high-speed aircrafts in the shape of :

- Power-assisted controls
- Fully powered controls

As the control forces is too high to be handled only by the pilot without any assistance



Power Assisted Controls



Power Assisted Controls

There is mechanical connection between the stick and control surface used in moderate speed and size aircrafts

When the pilot moves the stick, he deflects both :

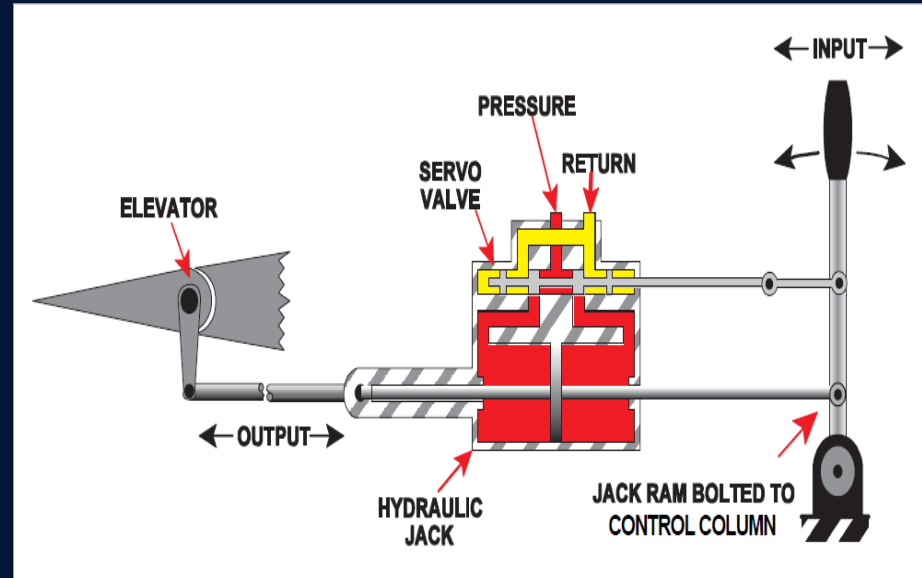
- Control surface
- Selector (servo) valve

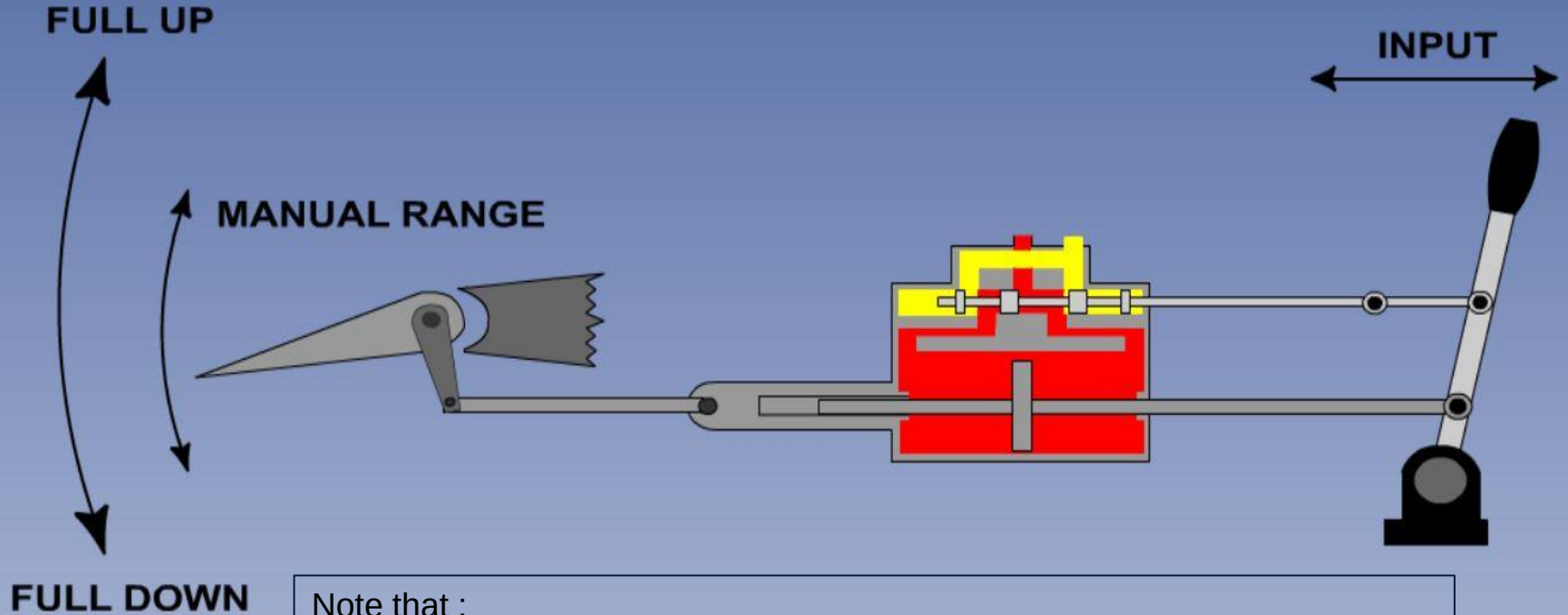
How it assists ?

- When the servo valve moves, it allows high pressure fluid to flow on one chamber of an actuator that also moves the control surface

Note! :

- Pilot still have the sense of the control surface (stick force)
- No need to design manual reversion system in case of hydraulic failure





Note that :

The piston is fixed, and the body of the actuator is free to move
So, the body moves until the selector (servo) valve is blocked (hydraulic lock) at that time the control surface will be in a new deflected angle according to the pilot input (new neutral position of the servo valve)

Fully Powered Controls

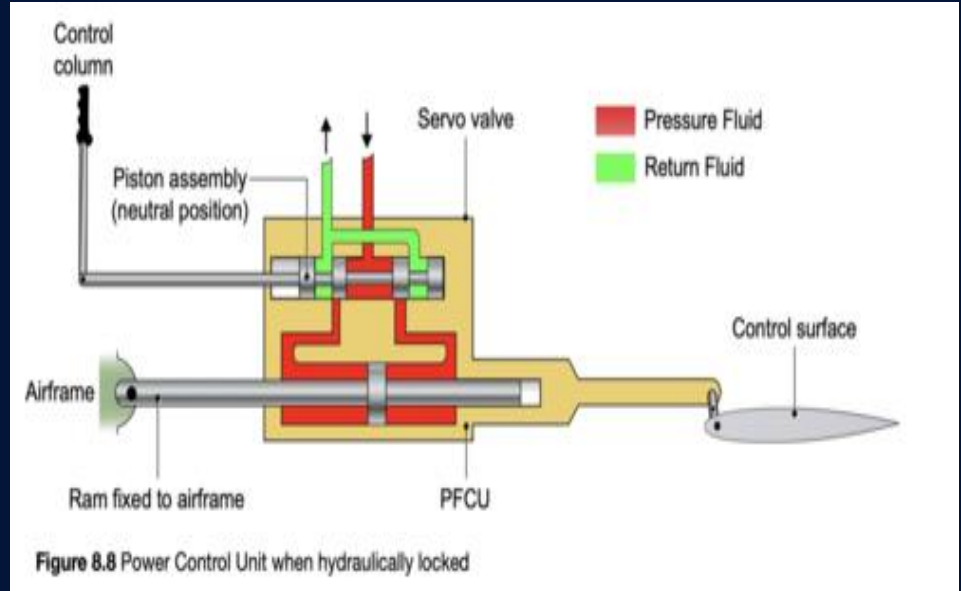
There is no mechanical connection (no stick force or feel)
Widely used in large size and high-speed aircrafts

How it works ?

- When the servo valve moves, it allows high pressure fluid to flow on one chamber of an actuator that also moves the control surface
- Deflection of the control surface is achieved only by the hydraulic actuator

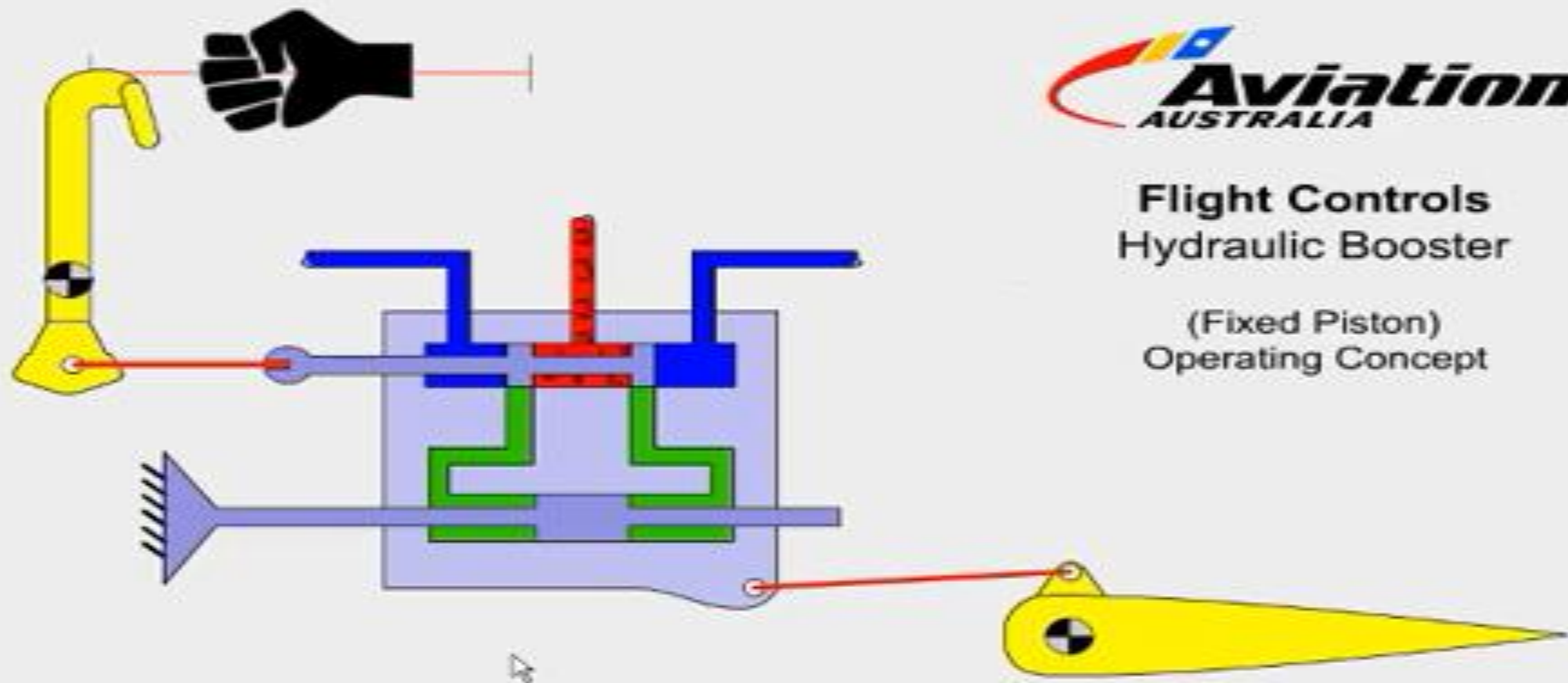
Note! :

We should give sense of the control force especially at high speed to the pilot by using artificial feel unit (Q-feel) otherwise he could accidentally overstress the aircraft



Flight Controls Hydraulic Booster

(Fixed Piston)
Operating Concept



INPUT:
Scroll Speed
1-10

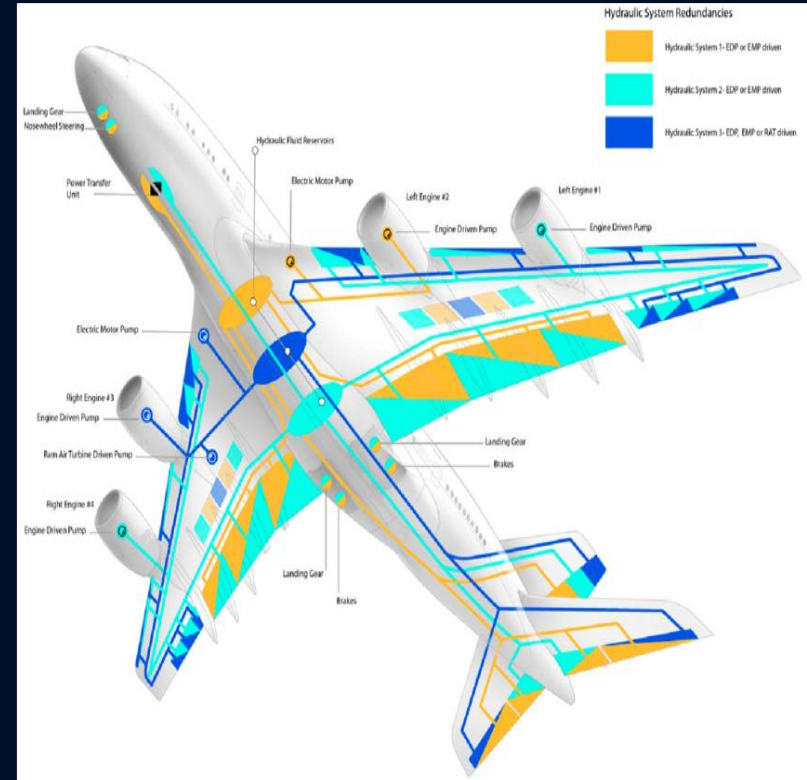
3

Fully Powered Controls redundancy

At all times, The control of aircraft is essential:

In case of hydraulic failure, the control of aircraft can be maintained by:

- Manual reversion using servo-tab or spring tabs.
- Use two or more separate independent power systems
- If all engine were lost, many aircrafts use RAT (ram air turbine) to provide power to the pumps to drive hydraulic fluid and keep the control system effective



Ram air turbine (RAT)



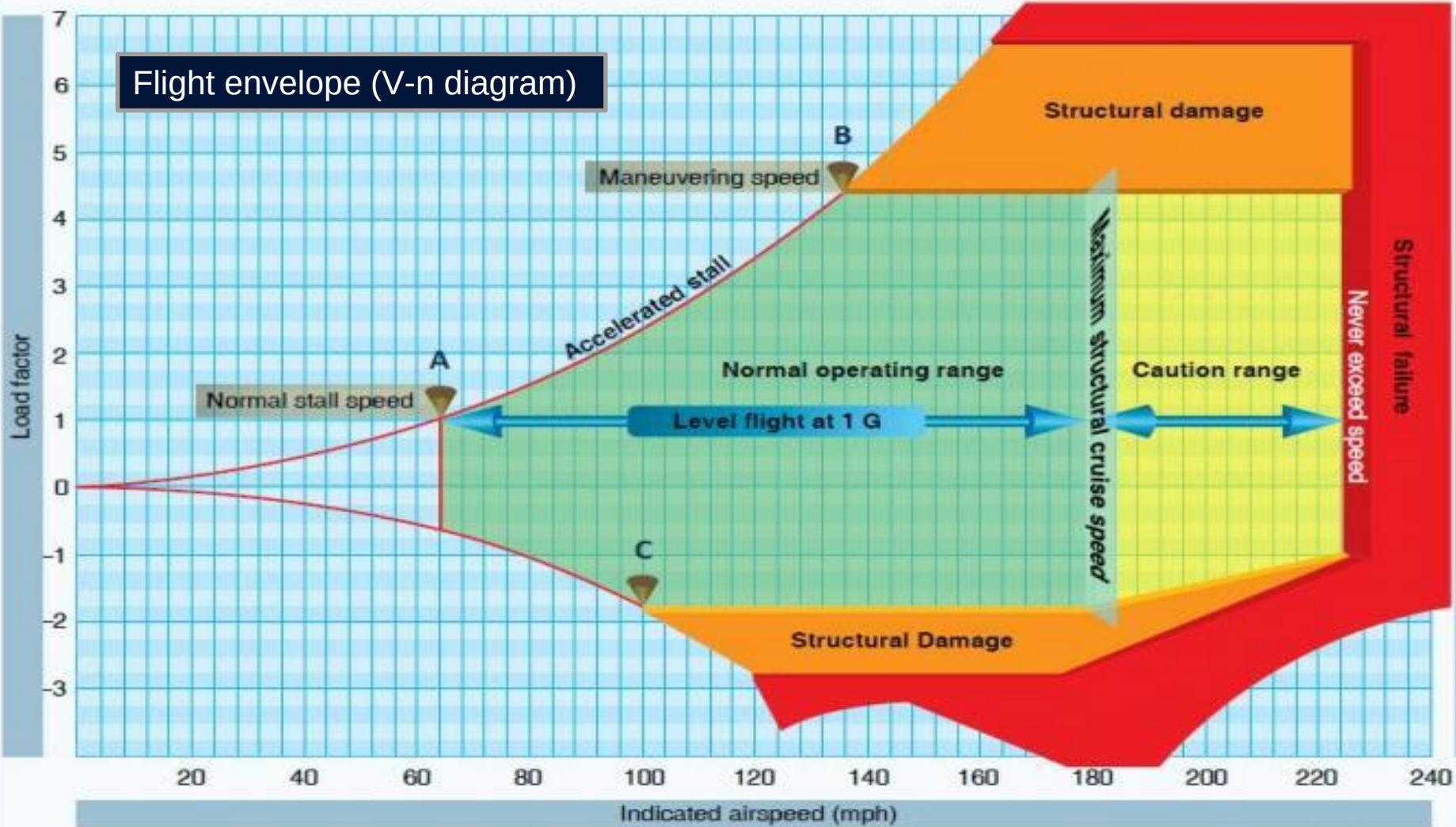


Flutter pheomena

How to avoid it ?!



Flight envelope (V-n diagram)

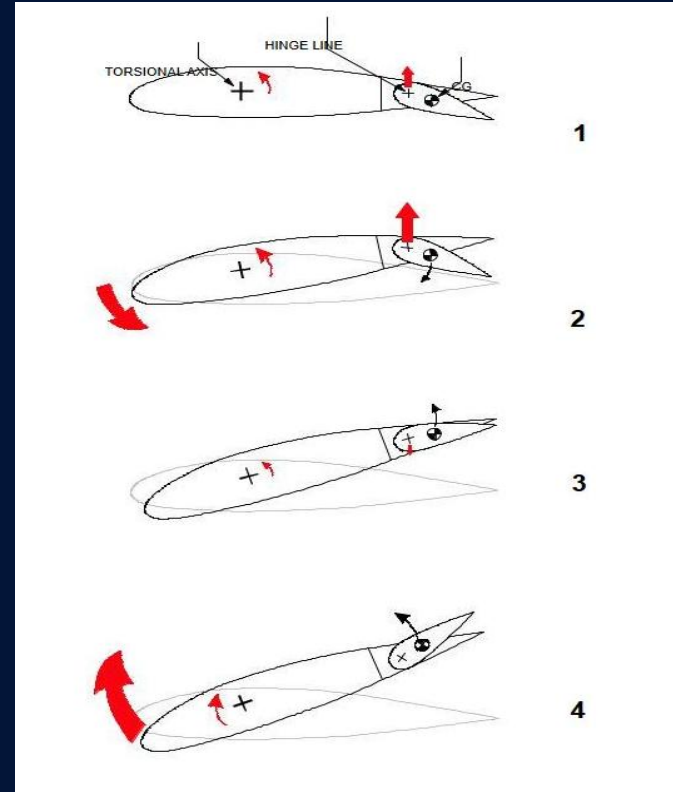


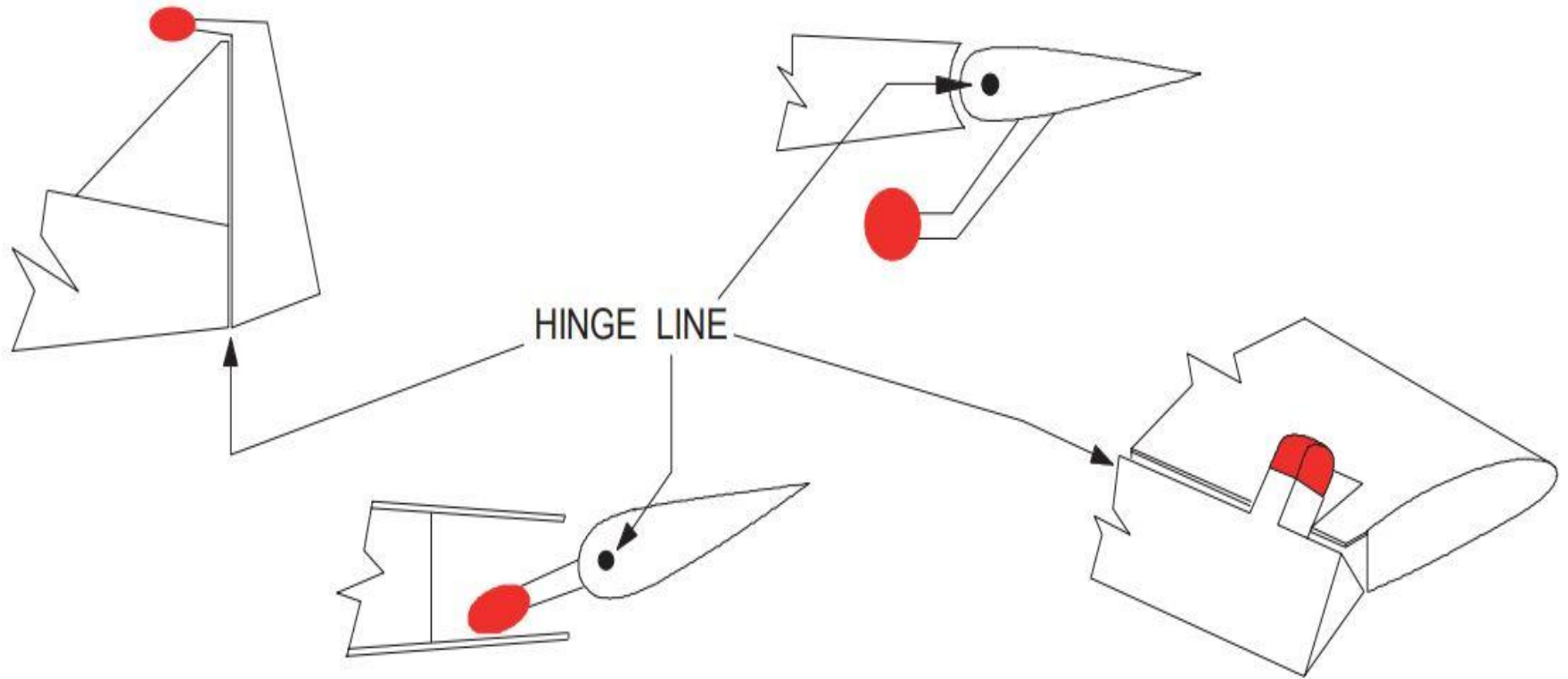
Flutter phenomena

- Flutter is a vibration phenomena which begins at the control surface when reaching the never exceed speed (V_{NE})
- It occurs when the CG of the control surface is behind the hinge line
- Flutter can not be controlled by the pilot
- Flutter must be avoided or prevented

How to avoid ?

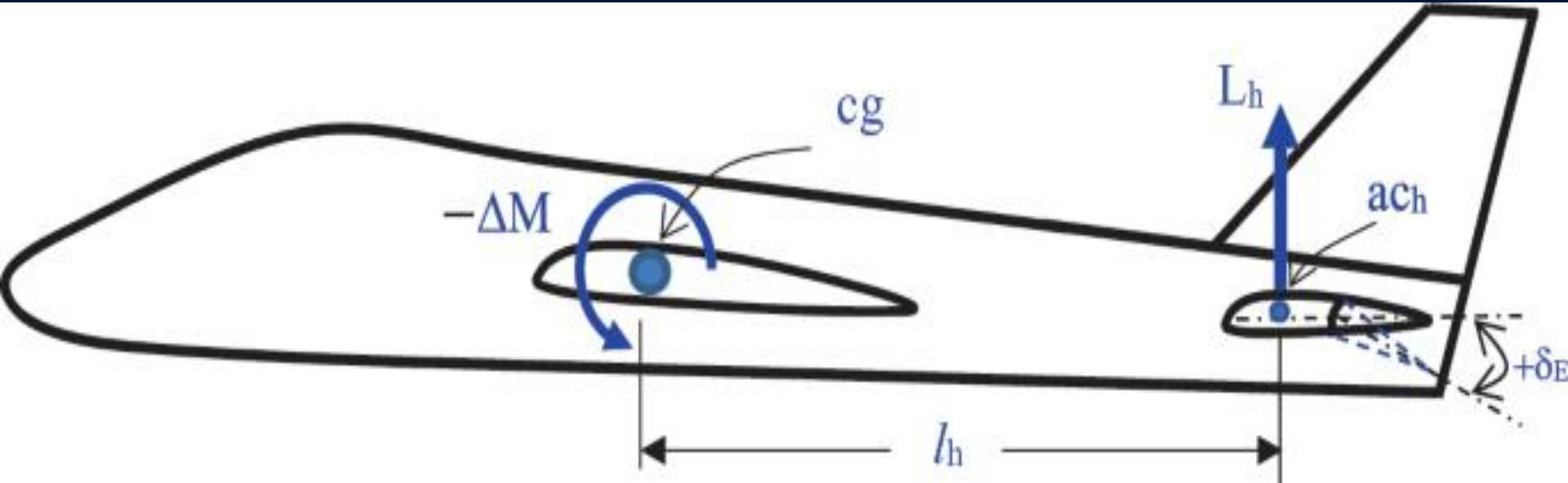
By adding weights to the control surface in front of the hinge line to make the CG of the control surface in front of hinge line that is what we call “mass balancing”
That would prevent flutter developing





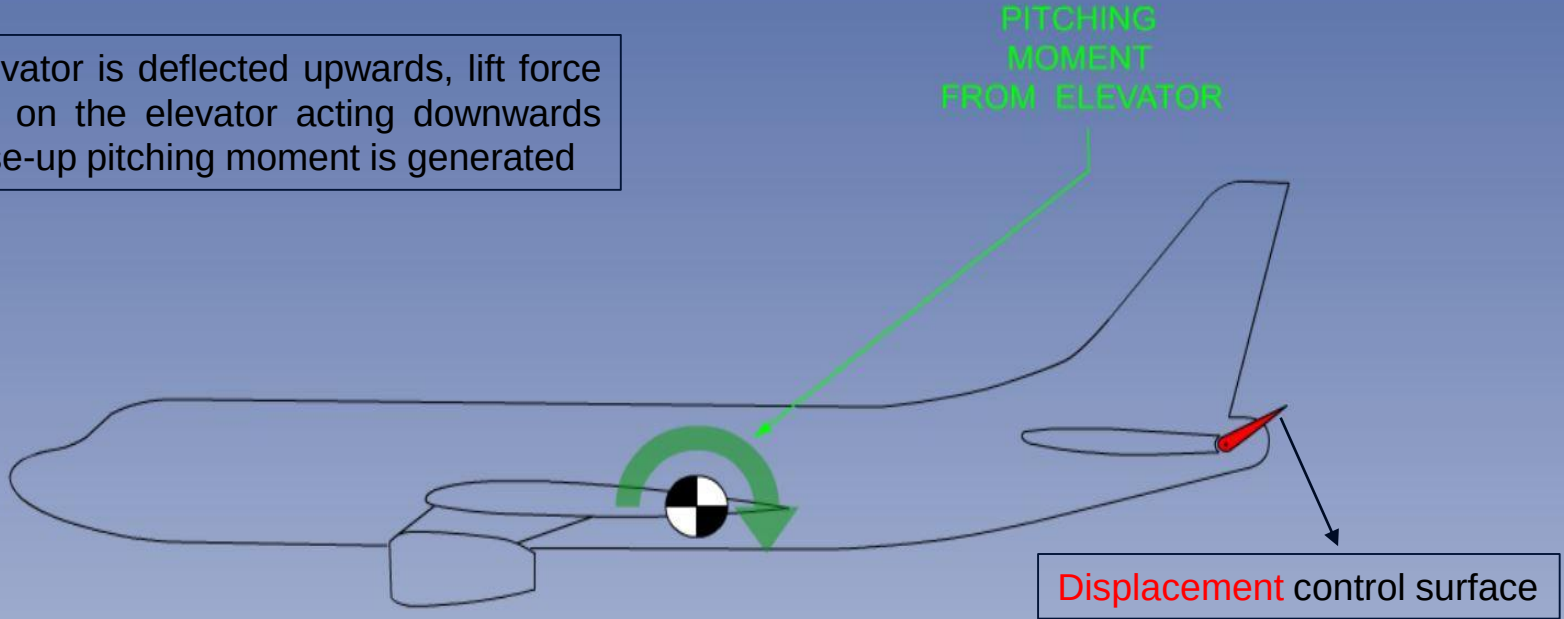
Note that:
the balancing weights are placed inside the aircraft structure to minimize the total drag

Longitudinal Control



Elevators or
all moving (slab) tailplane

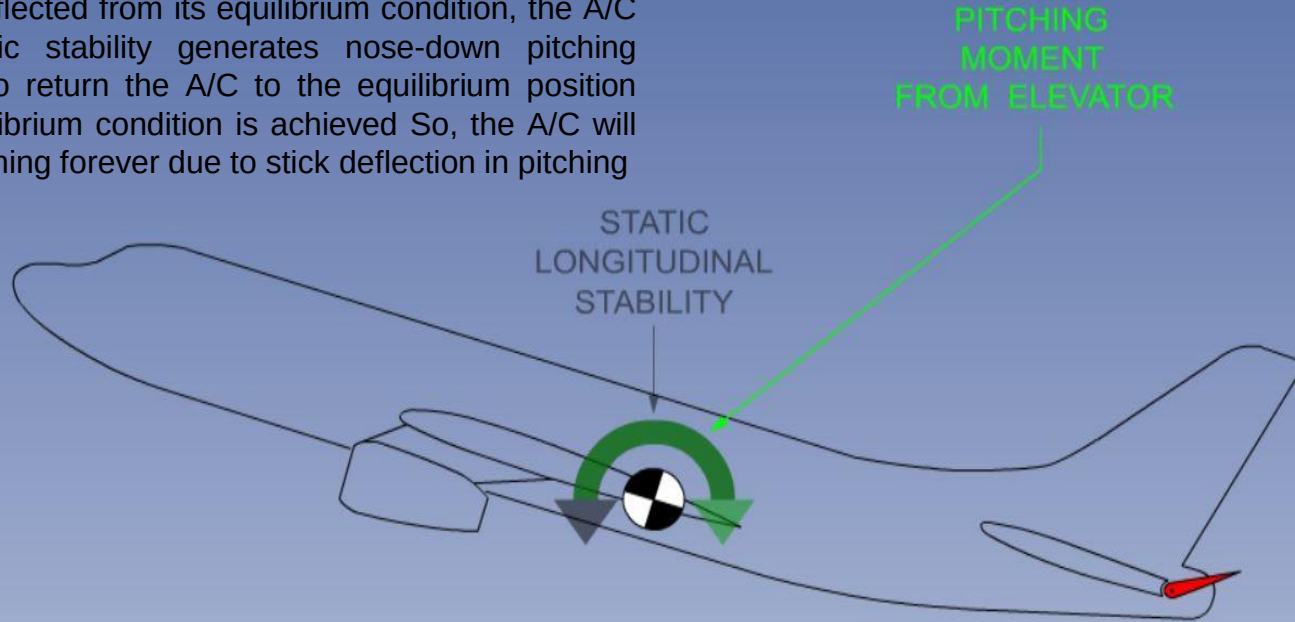
As the elevator is deflected upwards, lift force is created on the elevator acting downwards then a nose-up pitching moment is generated



But will the A/C continue pitching for infinity ?!

Elevators or all moving (slab) tailplane

As the A/C is deflected from its equilibrium condition, the A/C longitudinal static stability generates nose-down pitching moment trying to return the A/C to the equilibrium position until a new equilibrium condition is achieved. So, the A/C will not continue pitching forever due to stick deflection in pitching.



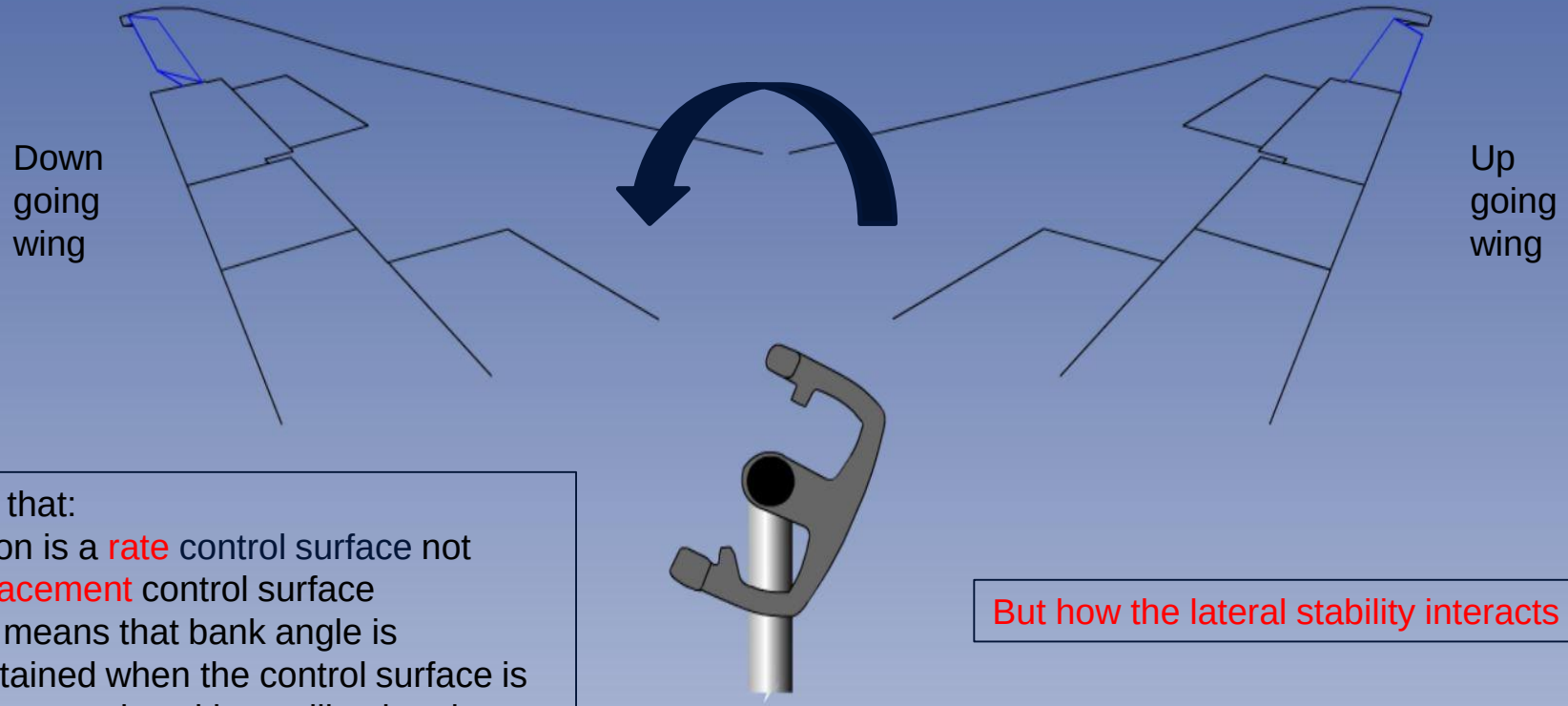
Nose-up pitching from elevator = Nose-down pitching from stabilizer

Note that: to keep on the new equilibrium condition, the stick must remain displaced from the neutral position

Lateral
control



Rolling Moment due to aileron deflection



Note that:
Aileron is a **rate** control surface not **displacement** control surface
That means that bank angle is maintained when the control surface is set to neutral position unlike the elevator

But how the lateral stability interacts ?!

Keeping
bank angle
with **neutral**
stick
position
after rolling

Espera y verás una de
RECONOCES
las fotos más raras
EL AVION??



Important notes on lateral control

- Aerodynamic damping in rolling decreases the rolling rate

for the same aileron deflection:

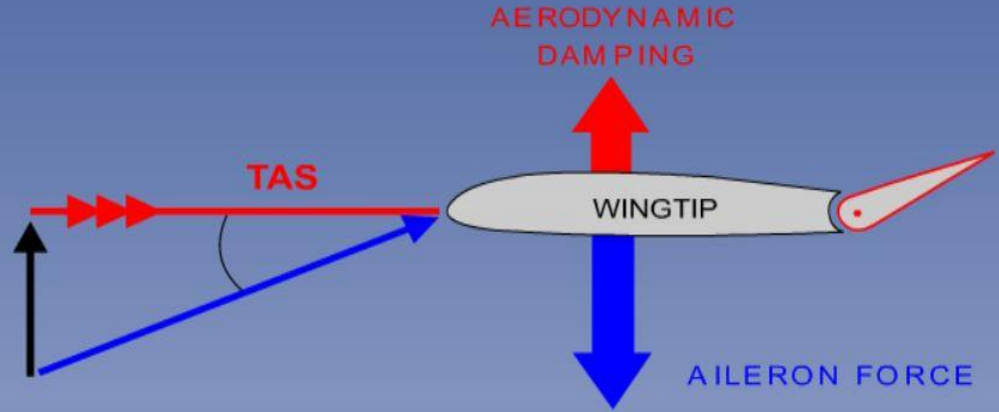
- As the wingspan increases, rolling damping increases
- as the true airspeed increases, rolling damping decreases

In rolling, the upgoing wing have more drag than the down going wing (adverse yaw) which can be avoided using:

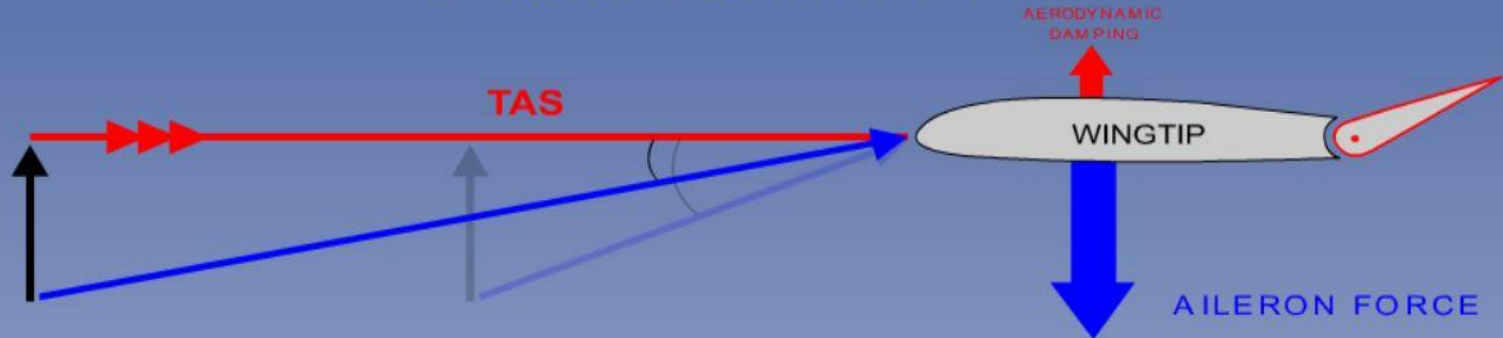
1. Differential aileron: upward aileron deflects more than downward aileron to increase drag on the down going wing
2. Frise aileron: the LE of the upgoing projects below the wing increasing parasite drag
3. Aileron-Rudder coupling : rolling control input automatically deflects rudder to cancel the adverse yaw
4. Using roll control spoilers



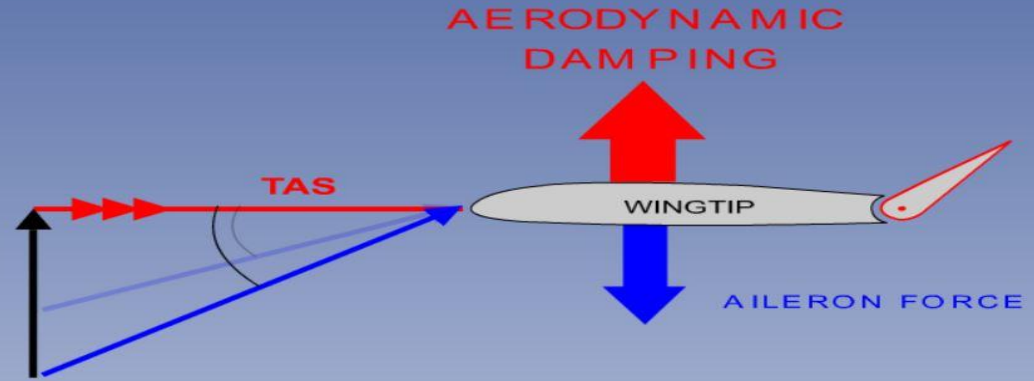
As the wing goes down,
there will be aerodynamic
force upwards and vice
versa in the up-going wing



INCREASED ALTITUDE



Increasing wingspan
Increases perpendicular relative airflow
Increasing the AOA
increasing the aerodynamic damping force



COMPARATIVE WING SPANS



Differential Ailerons

- usually differential and frise ailerons are used together in small and medium aircrafts



Up Aileron Deflects More Than Down Aileron



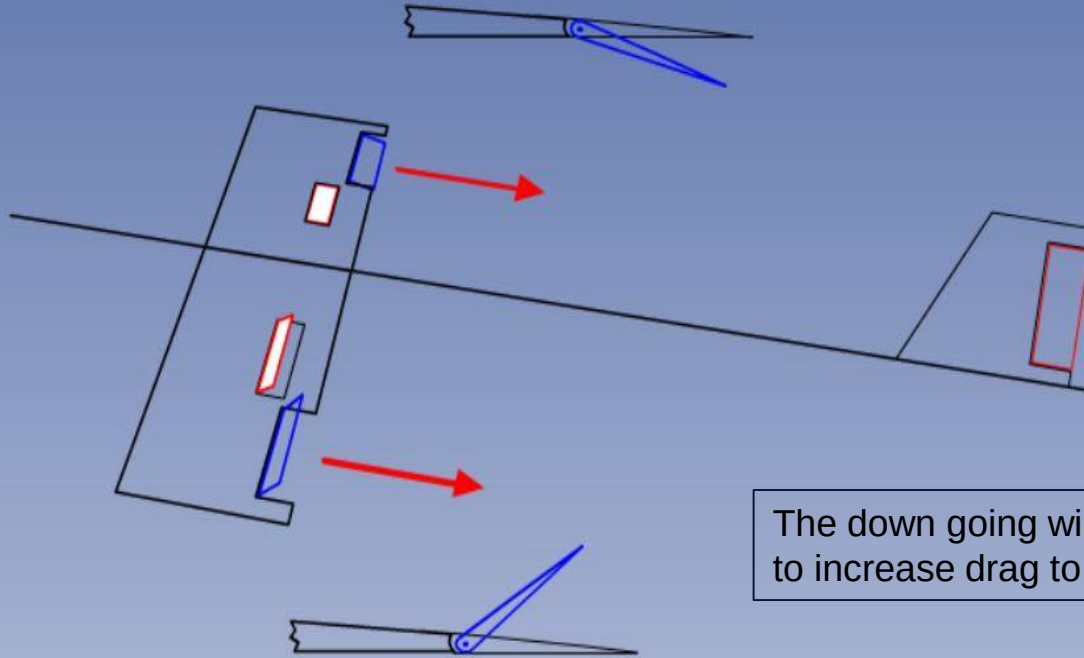
Frise Ailerons



**Bottom Of Up Aileron Pivots Into
Airstream, Creating Form Drag**

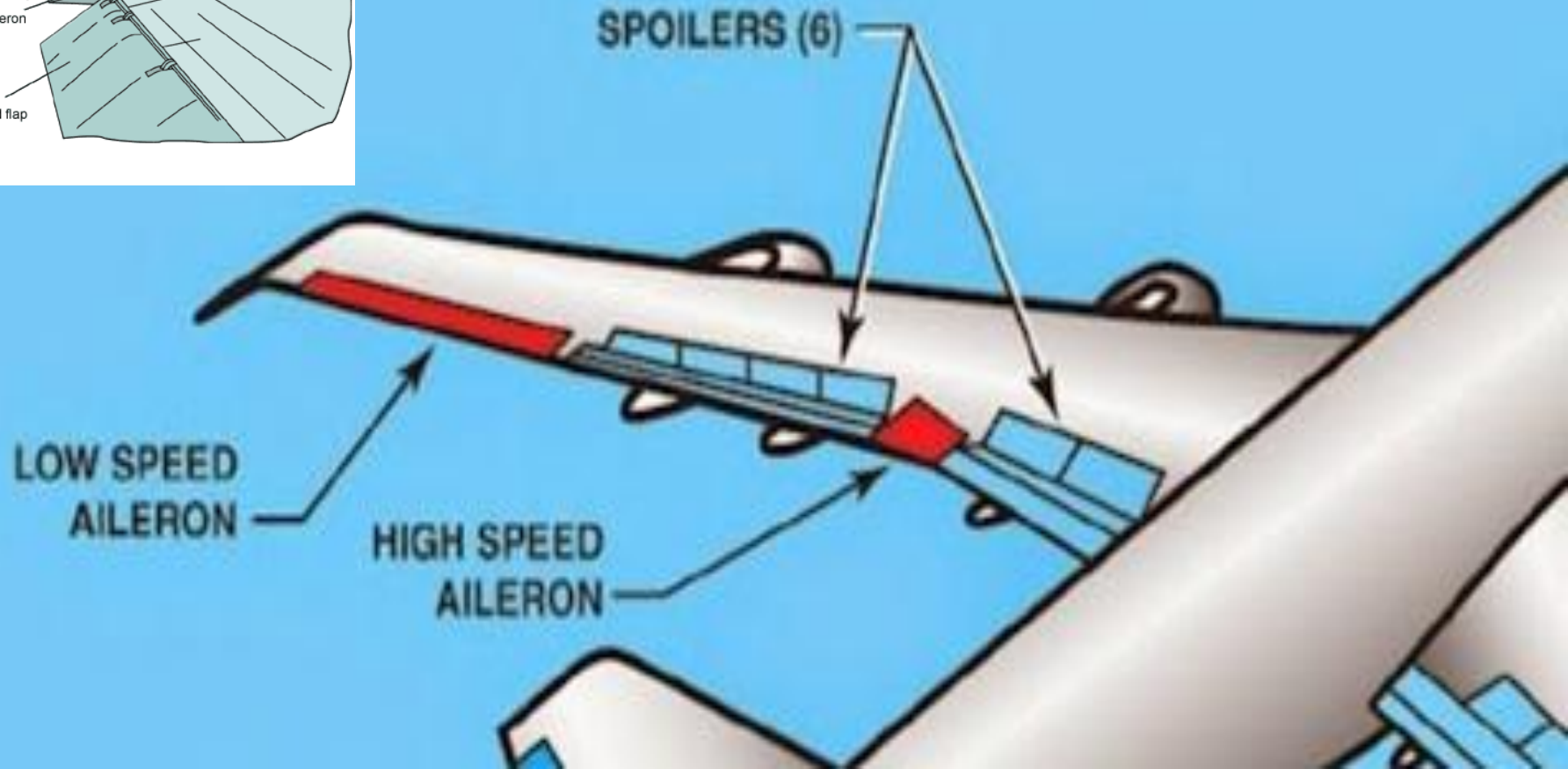
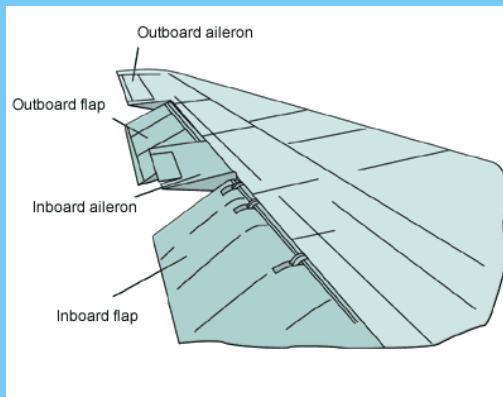
Modern, high speed jet transport aircraft usually use:

4. Roll Control Spoilers



The down going wing spoiler is deflected up to increase drag to cancel the adverse yaw

BOING 747 FLIGHT CONTROLS



Outboard ailerons vs inboard ailerons

- Outboard ailerons are put in the outboard section of the wing to increase rolling moment due to arm increase so it is very effective in low speeds

Disadvantage:

It maximize the chance of the aileron to make wing twist especially at high speeds as the wing twists downward at wing tip so the AOA would decrease and producing force that counter the effect of ailerons and decreasing the rolling moment

At very high speeds:

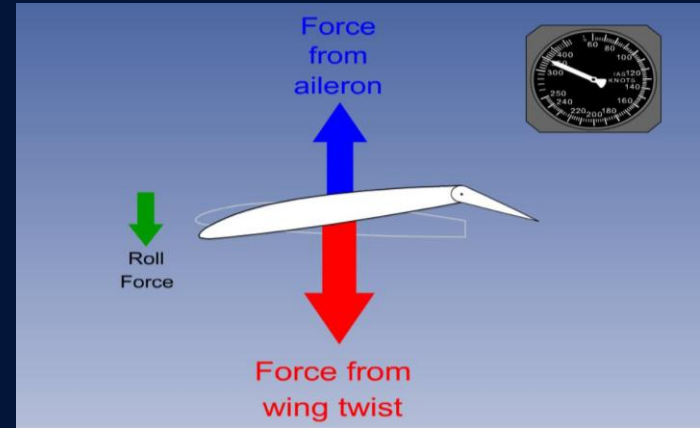
The rolling moment from the wing twist may be higher than that is produced by the aileron so the aircraft would roll on the opposite direction that was commanded by the pilot that phenomena is called

“ high speed aileron reversal “

Which limits the max operating speed of A/C

NOTE:

Even if aileron flutter is prevented by distributing masses at certain high speed control reversal will take place

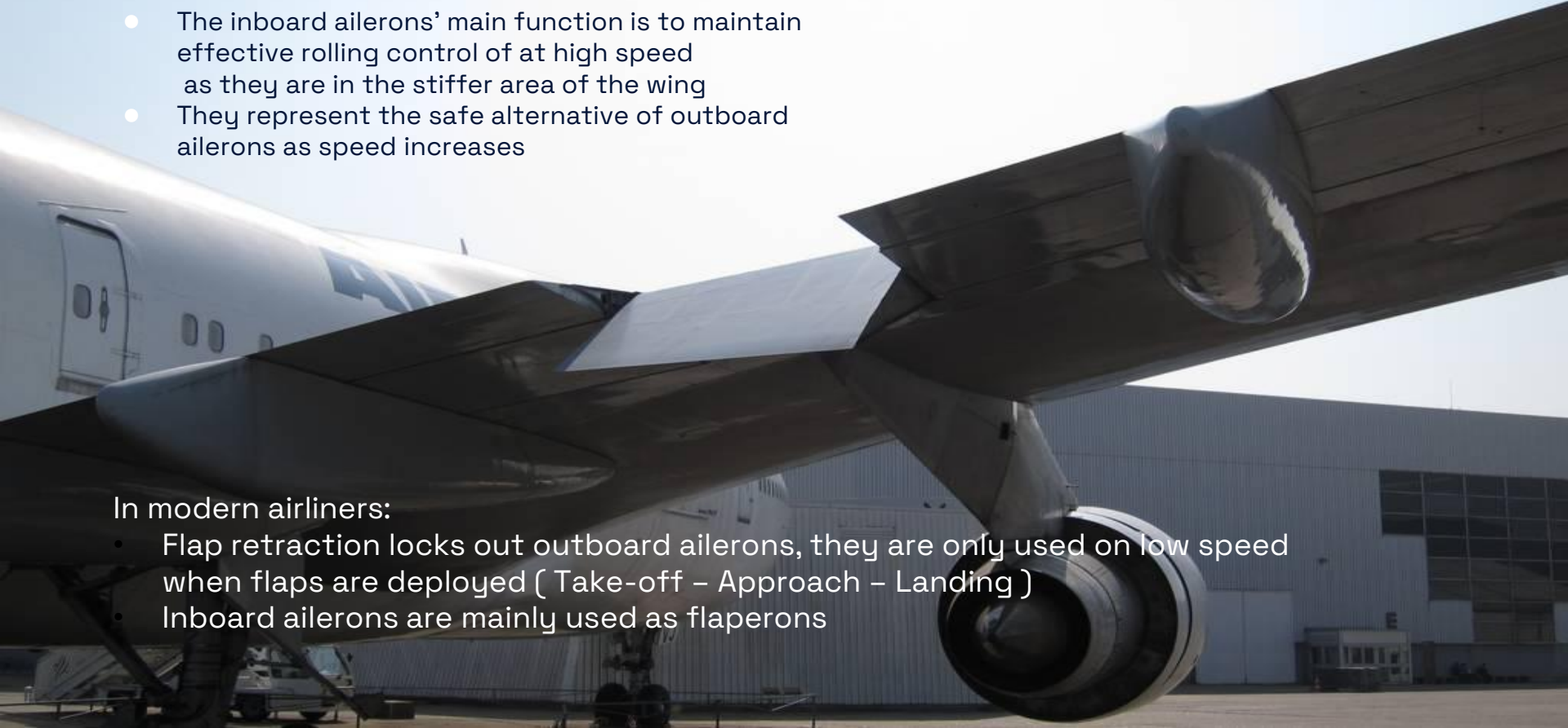


Outboard ailerons vs inboard ailerons

- The inboard ailerons' main function is to maintain effective rolling control of at high speed as they are in the stiffer area of the wing
- They represent the safe alternative of outboard ailerons as speed increases

In modern airliners:

- Flap retraction locks out outboard ailerons, they are only used on low speed when flaps are deployed (Take-off – Approach – Landing)
- Inboard ailerons are mainly used as flaperons



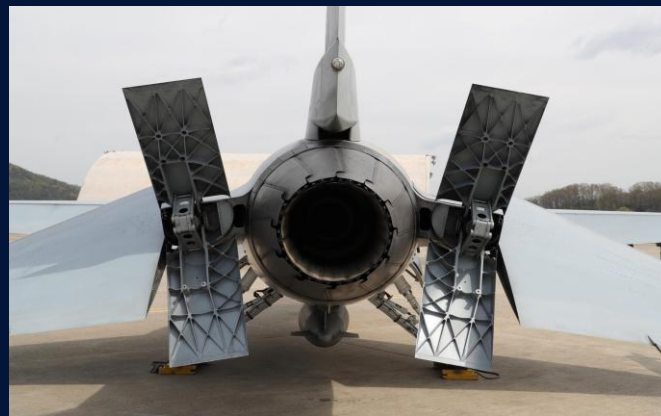
Speed brakes

There are two main types of airbrakes:

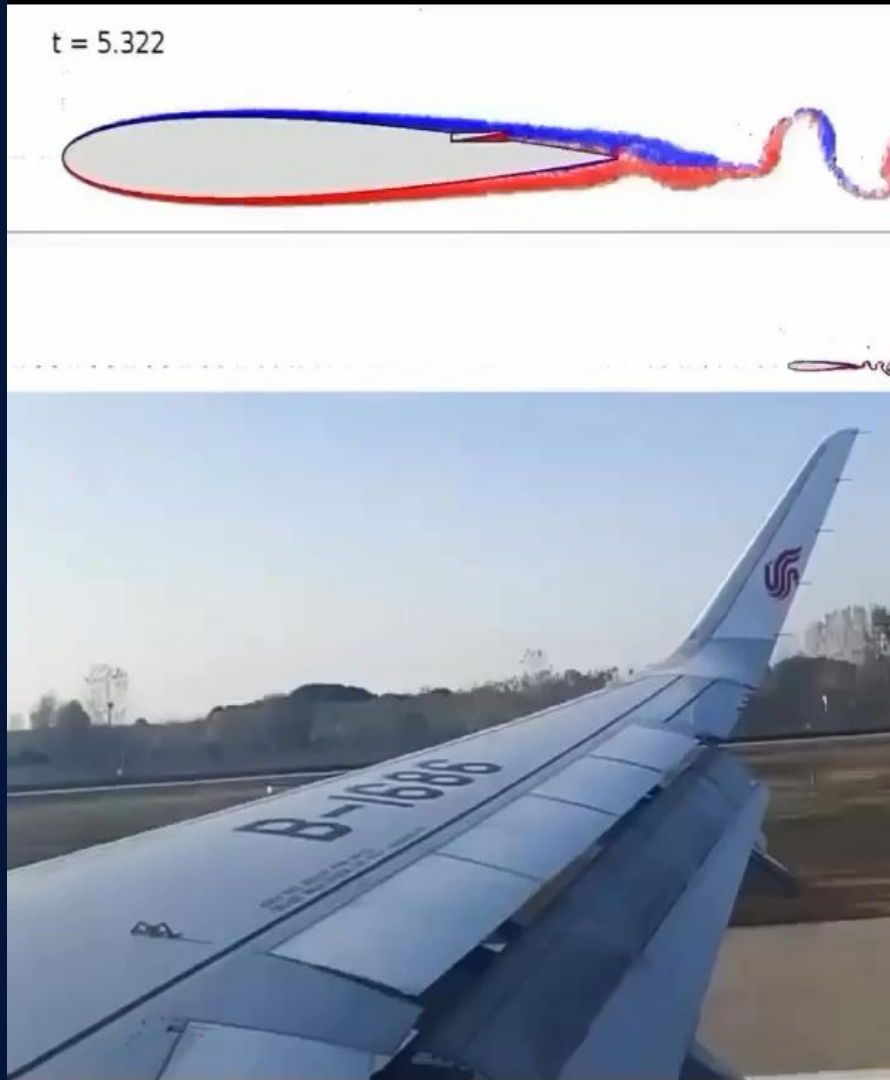
- Wing mounted speed brakes (spoilers)
- Rear fuselage mounted speed brakes

Note! :

- Spoiler speed brake angle of deflection depends on the IAS and lift restrictions
- There is a sensor that override any restriction in speed brake angle once the aircraft touches the ground and deploy the max deflection to brake the aircraft in the shortest distance and dumping any lift force “lift dump”



Spoilers as speed brakes



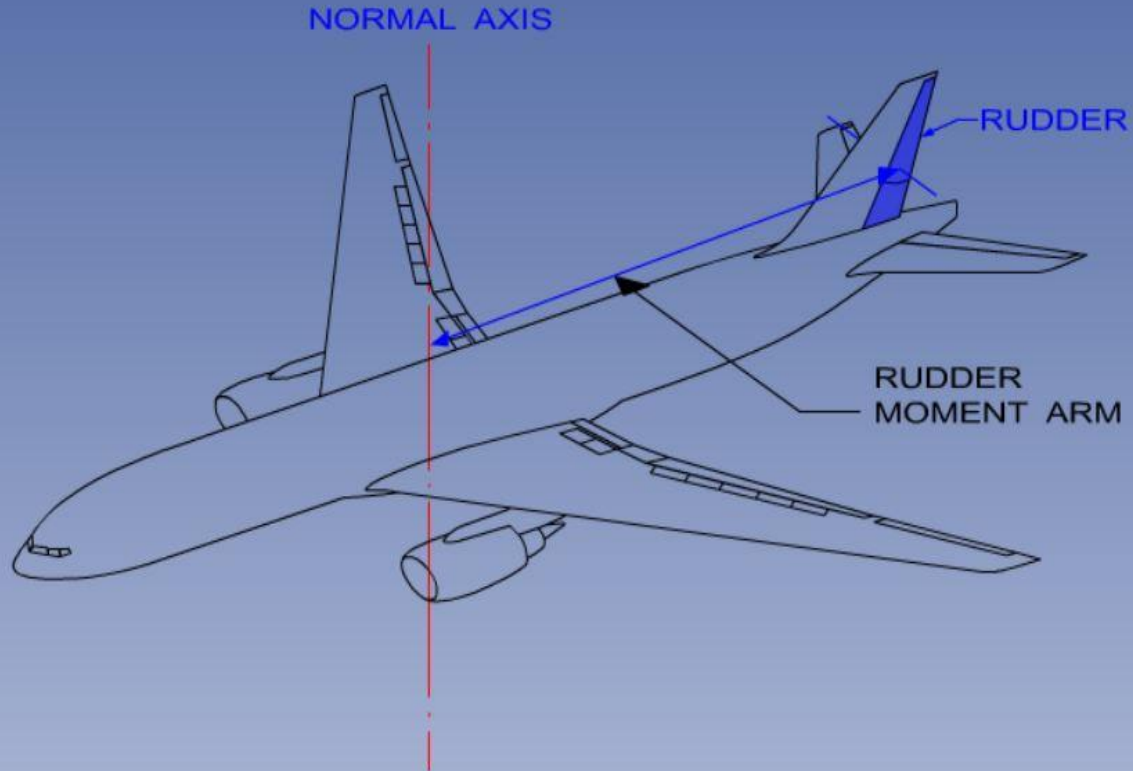
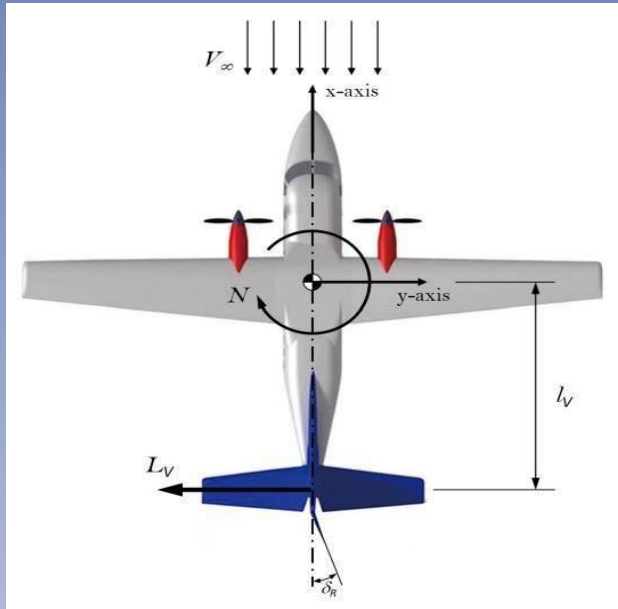
Directional control





RUDDER CONTROL

When rudder is deflected, lift force is generated on one side
Hence, generating yawing moment about the normal axis



Note:
Rudder is not used to steer the aircraft like a boat

Directional control

Main function of Directional Control (Rudder) :

- Balancing A/C by maintaining coordinated flight

Rudder also fulfil some extra functions :

- In case of **engine failure (asymmetric thrust)** :
it gives the pilot the ability to maintain directional control in multi-engine aircraft
- Take off or Landing on **cross wind**
- Balancing **adverse yaw** due to roll
- **Spin recovery**
- Opposing **propeller slipstream**



In case of engine failure:

In multi-engine aircrafts,

For example:

If the left engine is lost,

The A/C will yaw to the left

As the right wing is accelerated more than the left the A/C will experience roll to the left

As the lift force is tilted due to roll,

the A/C can go on a spiral dive

which can be catastrophic at low altitudes

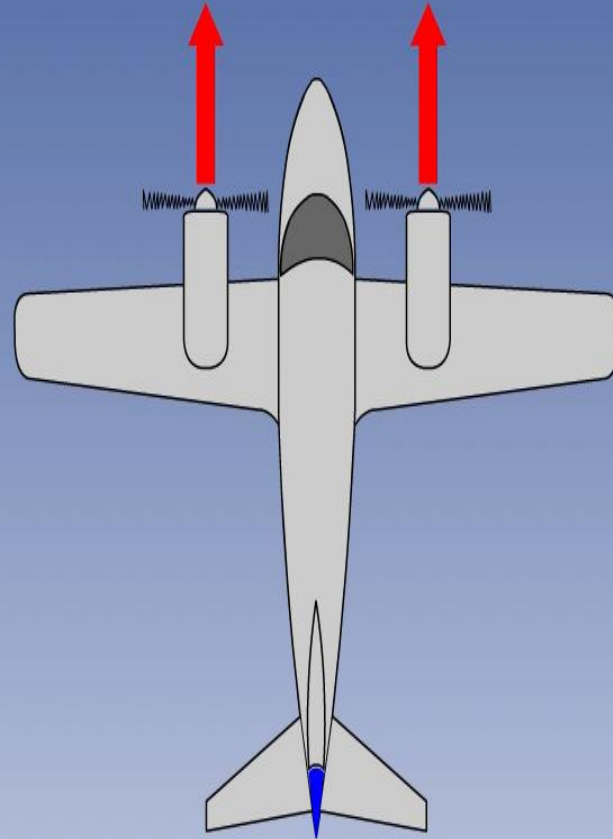
How to control this case?

The pilot should respond with rudder

correction to stop the yaw consequently

stop the roll and avoid

Going into spiral dive





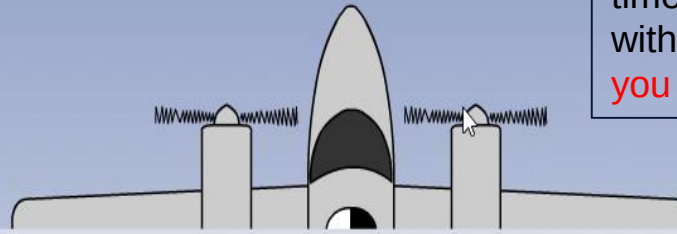
Directional Control

You should not use rudder to align the A/C with the runway centerline too early

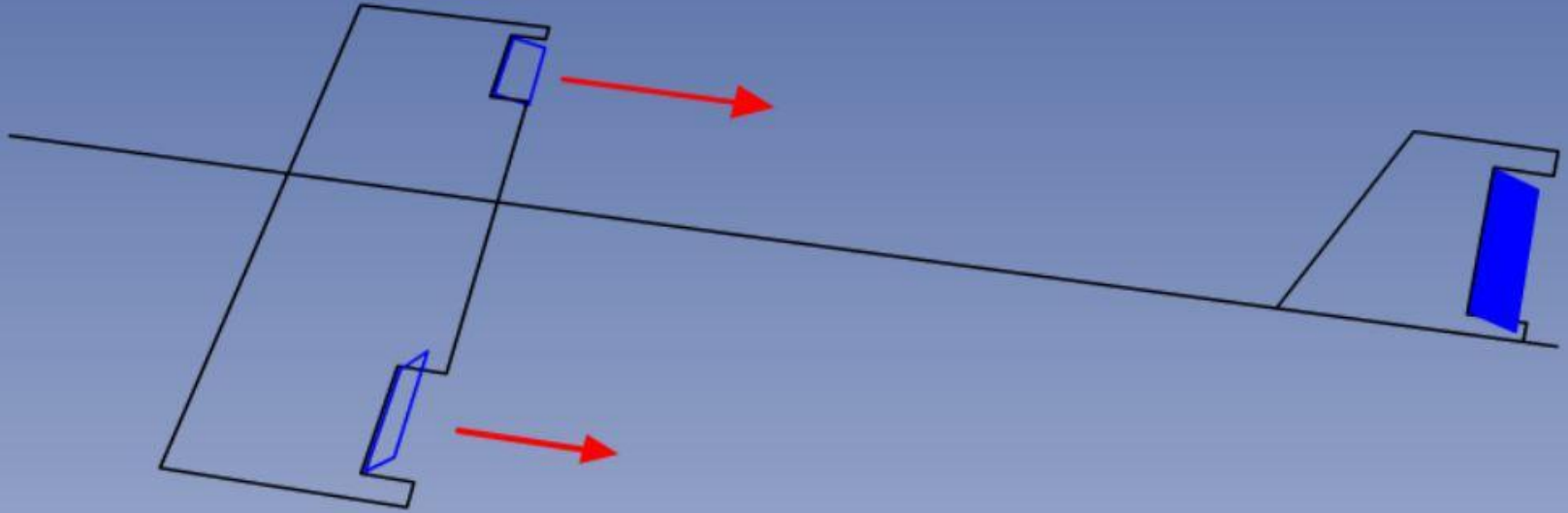
As the A/C will drift sideways



You should let the A/C weathercock the wind direction until the appropriate time then apply rudder to align the A/C with runway centerline *not too late or you will damage your LGs*



Balancing adverse yaw



Spin Recovery

P

POWER

OFF-CLOSE THROTTLE

**** NOW! IDENTIFY THE DIRECTION OF SPIN USING
OUTSIDE VIEW AND TURN COORDINATOR**

A

AILERONS

NEUTRAL

R

RUDDER

FULL OPPOSITE
TO DIRECTION OF SPIN
& 'PAUSE'

E

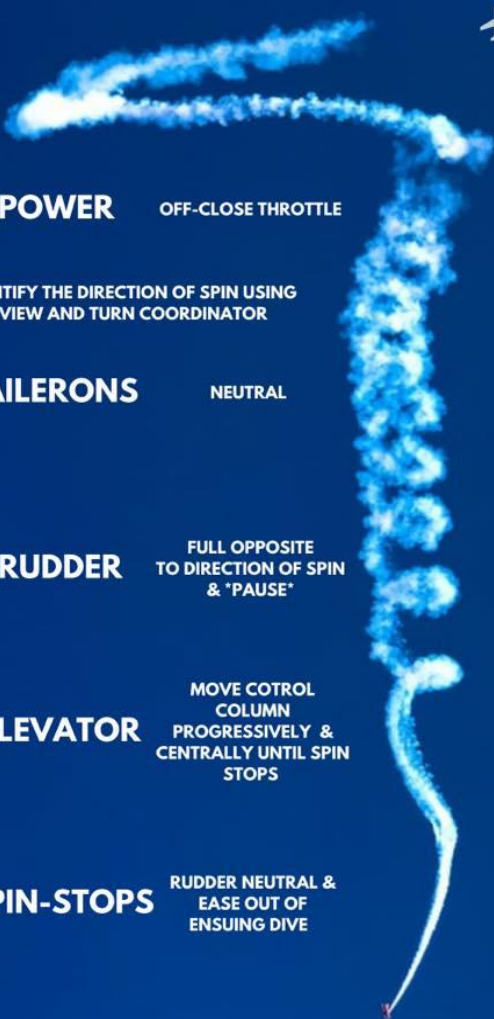
ELEVATOR

MOVE CONTROL
COLUMN
PROGRESSIVELY &
CENTRALLY UNTIL SPIN
STOPS

S

SPIN-STOPS

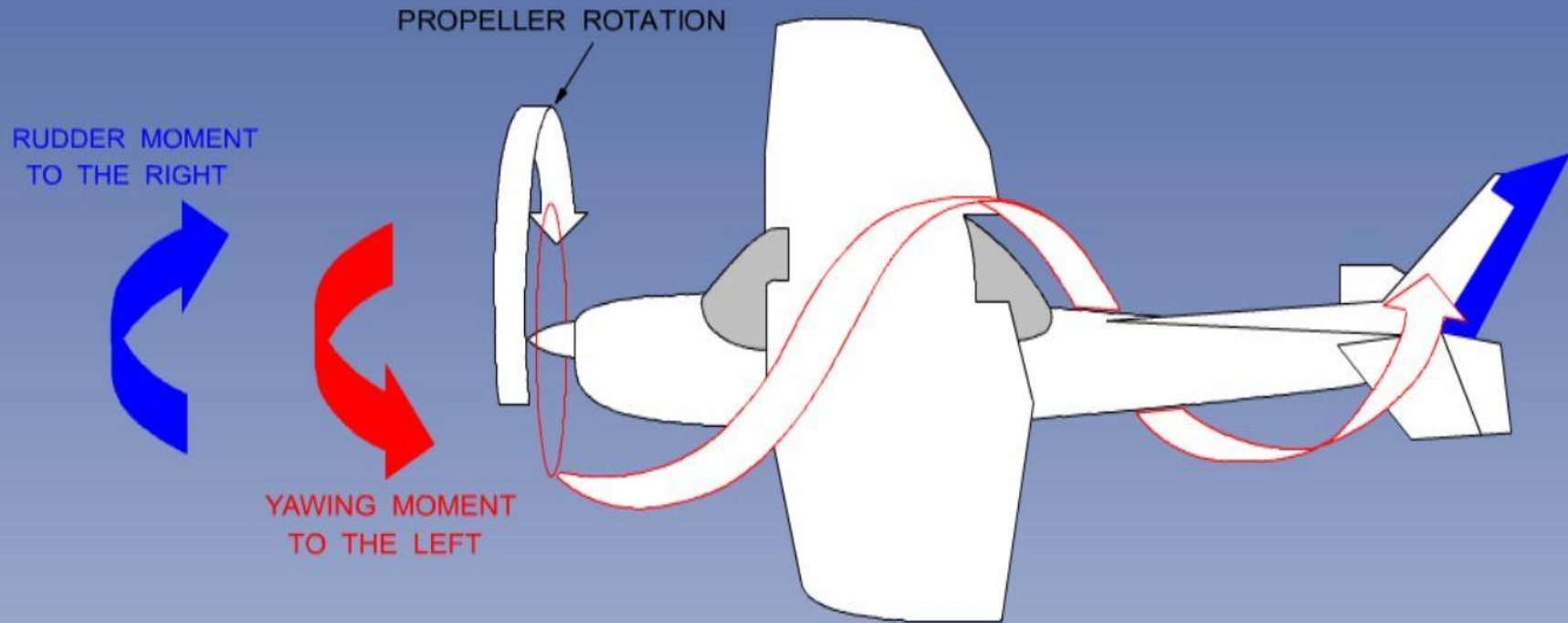
RUDDER NEUTRAL &
EASE OUT OF
ENSUING DIVE



How to recover ?

By applying the **PARE rule** by setting:
Power to idle
Ailerons to Neutral
Rudder full opposite the spin
Elevator smoothly forward





Slipstream

Slipstream causes:

- ° During a CLIMB & TAKEOFF

Yaw left (because of high thrust and high slipstream)

RIGHT RUDDER IS REQUIRED

- ° During a LEVEL segment

No yaw (Fixed rudder trim)

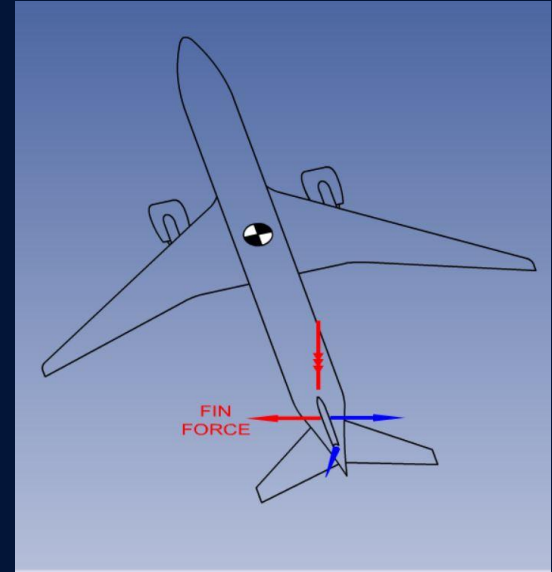
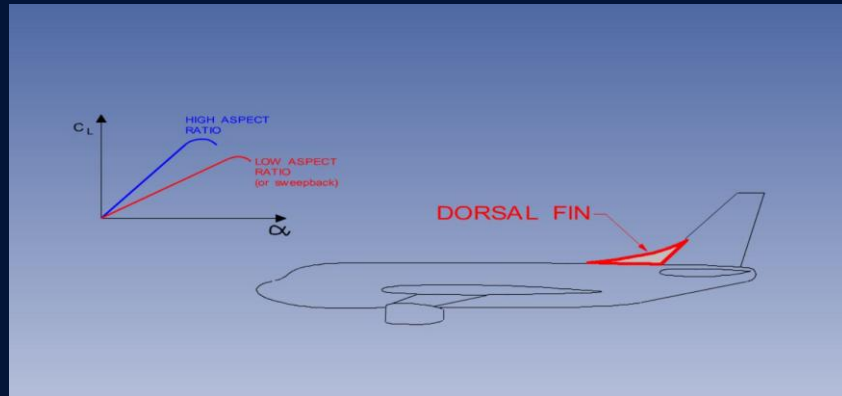
- ° During a DESCENT

Yaw right (No thrust and trim is fixed for cruise)

LEFT RUDDER IS REQUIRED

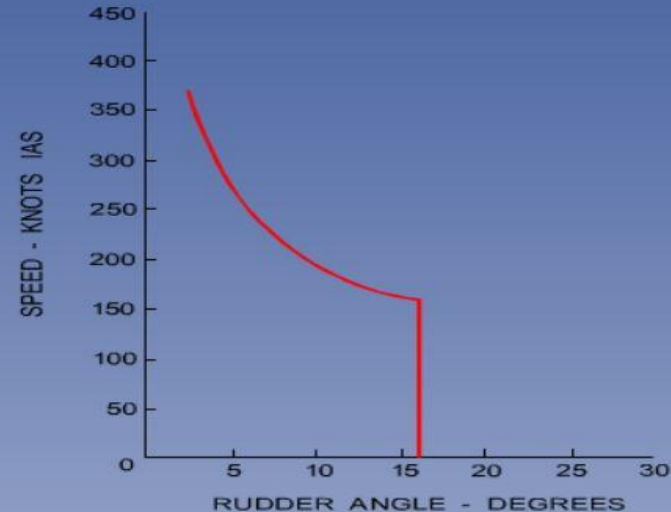
Notes on directional control

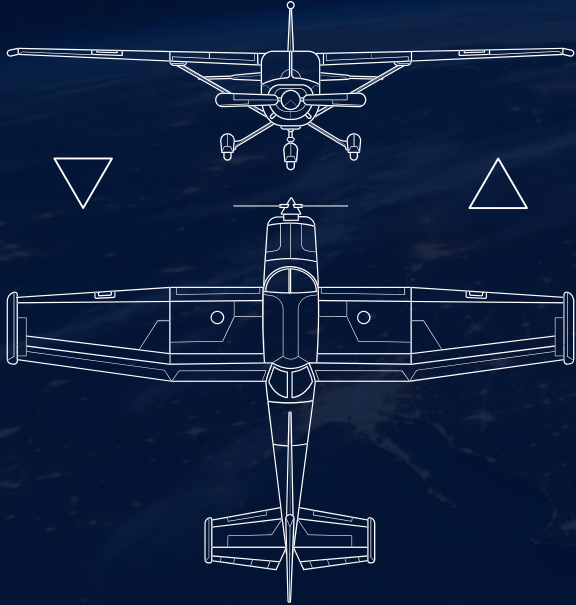
- Rudder is a displacement control surface
- At steady level flight :
the aircraft is said to be sideslipping at constant sideslip angle when the **fin force** = **rudder force** so the A/C will stop yawing about the vertical axis
- If cross wind is too great fin could stall (rudder lock) as sideslip angle would be too high this could be alleviated by sweeping fin backwards or using ventral or dorsal fin



Notes on directional control

- Rudder should be powerful enough to decrease sideslip angle of A/C to prevent fin stall
- Consider the most critical scenario of **one engine failure at low IAS** :
as the speed is low, rudder force will be small (not powerful enough) to bring A/C to straight flight **So, rudder deflection should be high enough to balance the A/C**
- Very high rudder deflection is **not permitted** at high speeds
- **Rudder ratio changer** is used to change the full rudder deflection at max pedal input according to IAS





REPORT !

- WHAT IS THE SECONDARY EFFECT OF ROLL ?!
- WHAT IS THE SECONDARY EFFECT OF YAW ?!

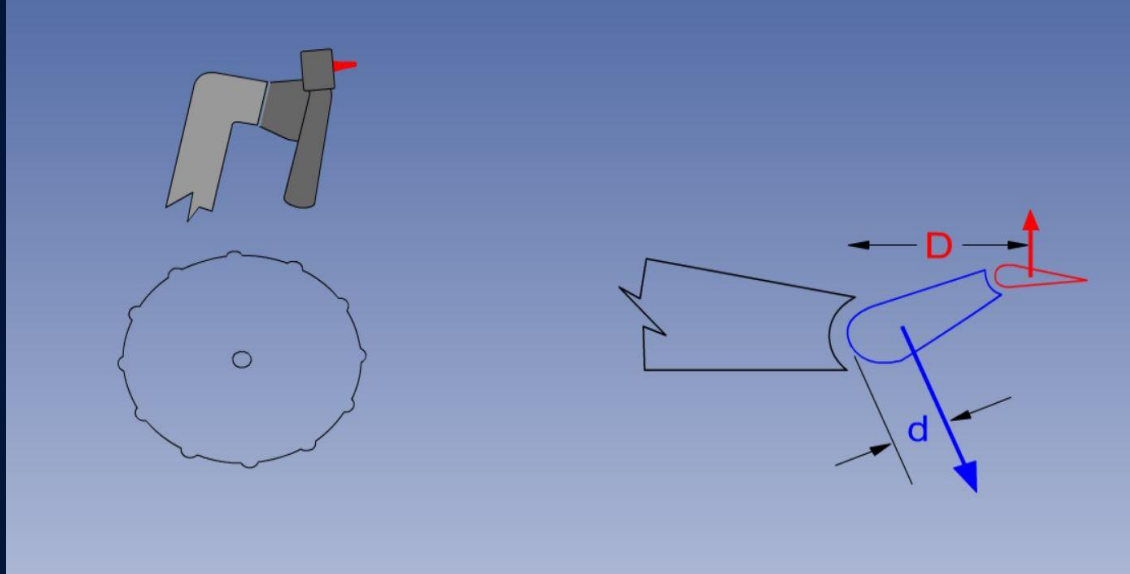


TRIMMING

(TRIM SYSTEMS)

WHAT is main function of trim system ?

- It enables the aircraft to maintain steady flight (Equilibrium/Trim condition) without any stick force or stick deflection required from the pilot
- This can be done by using trim tabs





Trim tab

Used in the majority of small aircrafts

How it works ?!

After deflecting the elevator to the required position to get the state of equilibrium, we adjust trim wheel/switch until the total hinge moment (stick force) is equal to zero

Keep in mind:

You need to keep the same deflection of the stick, but the stick force is zero

Disadvantages:

1. Reducing pitch authority

For stable (forward CG) A/C, we need high elevator deflection to trim the aircraft so there will be less pitch deflection available consequently, it will be hard to flare the A/C in landing

Variable incidence trimming tailplane

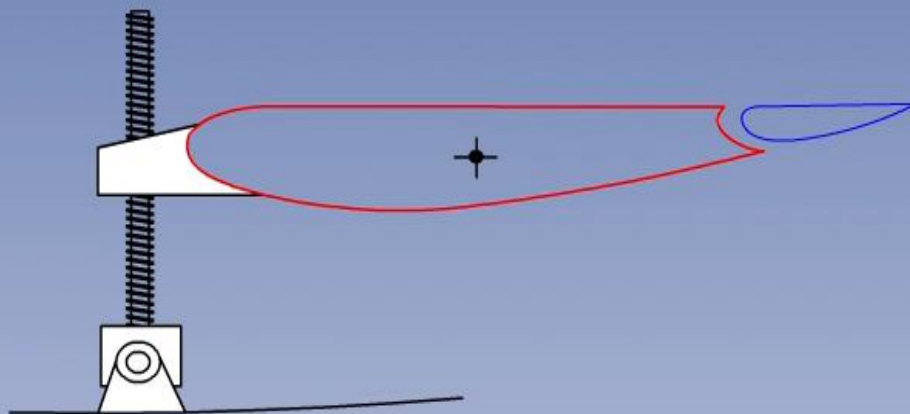
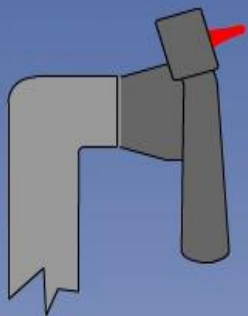
- For high speed, forward CG aircrafts trim tabs is not efficient as it does not provide the pitch trim authority required.
- So, **Variable incidence horizontal stabilizer** is widely used in modern jet transport

How it works ?!

The incidence of the stabilizer is changed by **screw jack** mounted on the rear fuselage and driven by motors from the gearbox at the base of the screw

- The pilot deflects the elevator to trim the A/C (steady flight)
- Rotate the trim wheel/ moving trim switches to change the incidence of the stabilizer
- While the stabilizer is moving (trimming), the elevator is being returned to neutral
- Zero stick force + stick is returned to neutral position unlike the trim tab (adv) consequently, we have the full pitch control capacity in all trim conditions
- Much more powerful and can trim high speed forward CG A/Cs







Only_FanCowl

Airbus A321
Trimmable horizontal
stabilizer

Variable
incidence
stabilizer
trim
system